

WENRUI HUANG

CONTACT

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EDUCATION

Brown University Providence, RI
Ph.D. in Mathematics 2022 - 2027 (*expected*)

- Advisor: Benoît Pausader

University of Science and Technology of China (USTC) Hefei, China
B.S. in Mathematics 2018 - 2022

- Advisor: Yong Wei
- Thesis: Mean curvature flow with surgery and its applications

RESEARCH INTERESTS

Partial differential equations, nonlinear dispersive PDEs and kinetic equations, stability and long-time behavior.

PAPERS AND PREPRINTS

1. *Scattering map for the Vlasov-Poisson system with a repulsive harmonic potential*, with H. Kwon. arXiv:2602.21344.
2. *Stability and instability of small BGK waves*, with D. Bian, E. Grenier, and B. Pausader. arXiv:2601.10030.
3. *The Vlasov-Poisson system with a perfectly conducting wall: Convex domains*, with B. Pausader and M. Suzuki. To appear in **Comm. Math. Phys.**
4. *Modified wave operators for the Hartree equation with repulsive Coulomb potential*, arXiv:2411.02737.
5. *Scattering of the Vlasov-Riesz system in three dimensions*, with H. Kwon. arXiv:2407.16919.

TALKS

- GLESPA Seminar, Brown University Apr. 2026
- Poster Session, Workshop on Kinetic Theory and Fluids, University of Wisconsin–Madison Mar. 2025
- Brown–Yale PDE Seminar Apr. 2024
- GLESPA Seminar, Brown University Apr. 2024
- Brown–Yale PDE Seminar Nov. 2023

TEACHING

Brown University

- TA, MATH 0180 Calculus III, Fall 2026
- TA, MATH 0190 Calculus II (Physics/Engineering), Fall 2025
- TA, MATH 0180 Calculus III, Spring 2024
- TA, MATH 0180 Calculus III, Fall 2023
- TA, MATH 0100 Calculus II, Spring 2023

University of Science and Technology of China

- TA, Riemannian Geometry, Spring 2022
- TA, Differential Geometry, Fall 2021
- TA, Topology, Spring 2021 and Spring 2022